

DB-Sorter

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1 Findings

- main algorithm is

$$H_{k+1} = e^{s[D, H_k]} H_k e^{-s[D, H_k]}$$

where D is a diagonal with sorted diagonal elements

- Sorting characteristic of DB happens for the discrete case as well: if $c_k = \text{tr}(H_k D)$

$$c_{k+1} - c_k = \text{tr}(\| [H_k, D] \|^2) + \mathcal{O}(s^2) \geq 0 + \mathcal{O}(s^2)$$

- each step gets closer to the ground energy:

$$\frac{d}{dt} \langle 0 | H(t) | 0 \rangle = 2 \langle \psi | d_0 I - D | \psi \rangle \leq 0$$

- in general for the k 'th excited energy level E_k we can show the following which does not guarantee a result similar to above for any excited states other than the last one:

$$\frac{d}{dt} (\langle k | H(t) | k \rangle - E_k)^2 = 4 (\langle k | H | k \rangle - E_k) (\langle \psi_k | d_k I - D | \psi_k \rangle)$$

- If D 's spectral gap is Δ and we can write $|\psi\rangle = H|0\rangle = \alpha|0\rangle + \beta|0^\perp\rangle$, then:

$$\frac{d}{dt} \langle 0 | H | 0 \rangle \leq -2\beta^2 \Delta$$

- Similar results can be achieved for the discrete case mostly with $\mathcal{O}(s^2)$ error
- for the case where H has standard basis as its eigenstates, we can make D diagonal in another basis
- using the above result and writing D in $|v_k\rangle = H^{\otimes n} |k\rangle$ with only $d_0 = -\pi$ and $s = 1$ we recover grover algorithm
- we can try different D 's to tradeoff number of two-qubit gates and convergence speed